Data Sources

State Greenhouse Gas Emissions Panel, 1997–2022

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`r Sys.Date()`

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Data Sources

This document lists every dataset used in this project, what it provides, and how to obtain it. Raw data is **not** stored in this repository because several files exceed GitHub's size limits. The cleaned versions **are** stored: data/interim/ holds one tidy file per source (state_fips, year, variable, value) and data/processed/ holds the final state-year panel. To reproduce the data from scratch, download each source below into data/raw/ and run the loaders in code/data_loaders/.

All sources are U.S. federal agencies, so the data is in the public domain.

Overview

Role	Source	Agency	Years used	Loader
Response	State GHG Inventory (gross emissions)	U.S. EPA	1997–2022	load_epa.py
Response (robustness)	SEDS (energy-related CO_2)	EIA	1997–2022	load_eia.py
Response (robustness)	GHG Reporting Program	U.S. EPA	2010–2022	not yet built

Role	Source	Agency	Years used	Loader
Control	nCLIMDIV heating and cooling degree days	NOAA NCEI	1997–2022	load_noaa.py
Control	SEDS (population, real GDP)	EIA	1997–2022	load_eia.py
Corporate	SEDS (fossil fuel production, grid mix)	EIA	1997–2022	load_eia.py
Corporate	GDP by State and Industry (SAGDP2)	BEA	1997–2022	load_bea.py
Corporate	Aggregates time series (sand and gravel)	USGS	1997–2022	load_usgs_minerals.py
Citizen Behaviors	Highway Statistics, Tables VM-2 and PS-1	FHWA	1997–2022	load_fhwa.py
Citizen Behaviors	T-100 Segment (domestic)	BTS	1997–2022	load_bts.py
Citizen Behaviors	SEDS (residential electricity)	EIA	1997–2022	load_eia.py
Citizen Behaviors	Personal Consumption Expenditures by State (SAPCE2)	BEA	1997–2022	load_bea.py
Citizen Behaviors	Quick Stats (livestock inventory)	USDA NASS	1997–2022	load_usda.py
Spatial weights	TIGER/Line state boundaries (queen contiguity)	Census Bureau	2022 boundaries	build_spatial_weights.py
Secondary analysis	nCLIMDIV temperature anomalies	NOAA NCEI	1997–2022	load_noaa.py
Secondary analysis	MODIS MOD11A1 land surface temperature	NASA LP DAAC	2001–2022	not yet built
Secondary analysis	NLCD land cover	MRLC	2015–2024	not yet built
Secondary analysis	Crop irrigation withdrawals	USGS	2000–2020	not yet built

Role	Source	Agency	Years used	Loader
Robustness only	FIA / EVALIDATOR	USDA Forest Service (USFS)	periodic	not yet built

Dropped Data

Role	Source	Agency	Reason
Corporate	Forms EIA-860 / EIA-923	EIA	Only 2025–2026 downloaded; SEDS already gives the state generation mix back to 1960
Corporate	Company Facts API (10-K CapEx)	SEC EDGAR	Firm-level national totals; can't be split by state, so year fixed effects would absorb it
Citizen Behaviors	Monthly Retail Trade	Census	National totals only; replaced by BEA state consumer spending (SAPCE2)

Every loader also reads data/state_crosswalk.csv (state name, postal code, FIPS code, NOAA state code), which is committed to the repository.

Response Variable

Pivot: The original response, year-over-year change in state temperature anomaly, was replaced with state gross greenhouse gas emissions per person. Greenhouse gases mix globally, so one state's emissions have almost no measurable effect on that same state's temperature in the same year, and year-over-year differencing removes the warming trend and leaves mostly weather noise. Emissions are causally linked to the drivers in both feature sets. Temperature is kept for a secondary, local-drivers analysis (see NOAA below).

U.S. EPA: Inventory of U.S. Greenhouse Gas Emissions and Sinks by State

- **Provides:** Annual state-level greenhouse gas emissions and removals, 1990–2022, by economic sector (residential, commercial, industry, transportation, electric power, agriculture, land use) and by gas (CO_2, CH_4, N_2O, fluorinated gases), in million metric tons of CO_2 equivalent. This gives us our main response variable, gross emissions per person, which excludes land use and forestry (LULUCF):

$$Y_{s,t} = \ln\left(\frac{\text{Gross GHG}_{s,t}}{\text{Population}_{s,t}}\right)$$

Net emissions (gross plus forest and land sinks) are kept as a robustness check. They are not the primary response because heavily forested states can come out near zero or negative (Maine is −0.3 MMT in 2022), which would mostly measure forest cover. - **Iron and steel breaks: how they are handled (build_panel.py).** Y itself is not edited. Three breaks are large relative to the state's gross emissions and get a flag and a step dummy:

Dummy	State-years	Iron and steel (MMT)	Change in Y
brk_mn_2010	Minnesota 2010–2022	0.08 → 17.57 (2009→2010)	+0.113 log points
brk_mi_2010	Michigan 2010–2022	2.01 → 12.00 (2009→2010)	+0.050
brk_md_2011	Maryland 2011–2012	0.17 → 4.67 → 0.00 (2010→2012)	−0.013, then −0.119

- y_break = 1 in any of those 28 state-years. The main model includes the three brk_ dummies as extra controls (3 parameters).
- Robustness Y: y_log_gross_noippu_pc = log((gross − ghg_ippu_mmt) / population), where ghg_ippu_mmt is EPA's whole IPCC sector "Industrial Processes and Product Use" (6.1% of 50-state gross in 2022, including ODS substitutes used by households and businesses). Without it, the 2010 change is −0.019 in Minnesota and −0.002 in Michigan, so the break is gone. Within the main sample it correlates 0.999 with Y.
- ghg_iron_steel_mmt (from load_epa.py) is kept in the panel so the break can be checked.
- Maryland's Sparrows Point mill closed in 2012, so the drop to zero is real. The 2011 spike is not.
- Indiana (12.7 → 19.8 MMT) and the KY/MO lime shifts are smaller relative to state gross and are not given dummies. The no-IPPU Y covers them.
- **Known breaks in state allocation:** Phase 3 found category-level breaks in the state series that look like changes in how EPA allocates emissions to states rather than real activity. The 50-state totals are continuous; only the split across states shifts.
  - *Iron and Steel Production, 2009→2010.* Emissions move from many states (OH 4.6→2.0, PA 4.3→1.8, AL 3.6→1.1, TN 2.6→0.2, TX, SC, WI, …) to a few (MN 0.08→17.6, MI 2.0→12.0, IN 12.7→19.8 MMT) and stay there. This is most likely the switch to facility-level GHGRP data from 2010 (not yet confirmed in EPA's methodology annex). Minnesota's gross emissions jump about 12% in 2010 (taconite pellet plants). Lime Production in KY (0.4→1.6) and MO (0.3→1.8) shows the same pattern. State and year fixed effects do not absorb a state-specific level shift.
  - *Fluorochemical Production, 1999→2000.* West Virginia is a flat 17.1 MMT in 1997–1999, then 0.5. AL, AR and KS also fall to near zero by 2005, consistent with the national decline in HFC-23 by-product emissions, but the flat early values suggest an extrapolated estimate.
  - Other one-year jumps over 10% (relative to the national median change) are real power-plant or oil-field events: for example, NV 2006 (Mohave plant closure), TN 2009, AR 2015, ND 2018 (Bakken petroleum systems), and DE 2009–2010 (Delaware City refinery idled late 2009, restarted 2011).
- **Citation:** U.S. Environmental Protection Agency. Inventory of U.S. Greenhouse Gas Emissions and Sinks by State: 1990–2022. Consolidated All-State GHG Data (AllStateGHGData90-22_v082924). Retrieved on Nov 2, 2026 from https://www.epa.gov/

ghgemissions/state-ghg-emissions-and-removals

U.S. EPA: Greenhouse Gas Reporting Program *(secondary analysis)*

- **Provides:** Facility-reported greenhouse gas data for large emitters (2023 data summary spreadsheets).
- **Citation:** U.S. Environmental Protection Agency. Greenhouse Gas Reporting System. 2023 Data Summary Spreadsheets. Retrieved on Oct 29, 2026 from https://www.epa.gov/ghgreporting/data-sets.

Controls and Secondary Analysis

NOAA NCEI Climate Divisional Database (nCLIMDIV)

- **Provides:** Monthly statewide temperature, heating degree days (HDD), cooling degree days (CDD), and precipitation, dating back over a century. HDD and CDD are controls for how much heating and cooling each year required. The temperature anomaly (each state's annual mean minus its own 1991–2020 normal) is the response for the secondary analysis:

$$\text{Anomaly}_{s,t} = \bar{T}_{s,t} - \bar{T}_{s,\,1991\text{–}2020}$$

- **Citation:** Vose, Russell S.; Applequist, Scott; Squires, Mike; Durre, Imke; Menne, Matthew J.; Williams, Claude N., Jr.; Fenimore, Chris; Gleason, Karin; Arndt, Derek (2014): NOAA Monthly U.S. Climate Divisional Database (NClimDiv). /data/nclimdiv-monthly/archive. NOAA National Climatic Data Center. doi:10.7289/V5M32STR. Oct 19, 2026.
- **Notes: Hawaii is not in nCLIMDIV, and nCLIMDIV gives Alaska temperature and precipitation but no heating or cooling degree days.** Because HDD and CDD are controls, the main model runs on the **contiguous 48 states (1,248 state-years)**. Alaska and Hawaii stay in the panel with missing HDD/CDD (in_main_sample = 0). HDD and CDD enter the models as hdd_k and cdd_k (thousands of degree-days) so their coefficients are readable.

No link currently available. The source is undergoing data migration and the metadata XML is unretrievable.

MODIS Thermal Data *(secondary analysis)*

- **Provides:** Land surface temperature and emissivity, used to derive state-level Urban Heat Island spatial index metrics for the secondary (local temperature) analysis.
- **Citation:** National Aeronautics and Space Administration (NASA) Land Processes Distributed Active Archive Center (LP DAAC), MODIS/061/MOD11A1 Land Surface Temperature and Emissivity Daily L3 Global 1km. Retrieved on Oct 19, 2026 from Google Earth Engine.

USGS Water Use Data and NLCD *(secondary analysis)*

- **Provides:** Secondary analysis only. These affect local temperature, not greenhouse gas emissions.

  - USGS Water Use in the United States: state-level irrigation withdrawal data, used to estimate irrigation density.
  - NLCD: classification of developed open space, high-density urban areas, and forest cover across every state.
- **Citation 1 (USGS):** Haynes, J.V., Read, A.L., Chan, A.Y., Martin, D.J., Regan, R.S., Henson, W.R., Niswonger, R.G., and Stewart, J.S., 2023, Monthly crop irrigation withdrawals and efficiencies by HUC12 watershed for years 2000-2020 within the conterminous United States (ver. 2.0, September 2024): U.S. Geological Survey data release, https://doi.org/10.5066/P9LGISUM. Retrieved on Oct 30, 2026.
- **Citation 2 (NLCD):** Multi-Resolution Land Characteristics Consortium. (2024). National Land Cover Database. Land Cover (CONUS), 2015-2024. Retrieved on Oct 30, 2026 from https://www.mrlc.gov/data.

Feature Set A: Corporate and Supply-Side Metrics

EIA: State Energy Data System (SEDS)

- **Provides:** Annual state-level fossil fuel production (coal, crude oil, natural gas), the fossil fuel share of electric power sector energy input, population, real GDP, and energy-related CO_2 emissions by sector (robustness response). Also transportation-sector CO_2 by fuel (co2_trans___mmt: motor gasoline, distillate, jet fuel, residual, aviation gasoline, HGL, lubricants, natural gas, coal), which sum to transport CO_2 within 0.003 MMT. These are used **only** to split transport emissions in the descriptive decomposition; they are fuel consumption and never enter a treatment.
- **Citation:** U.S. Energy Information Administration. (2026). State Energy Data System. Consolidated data file, Consumption, price, expenditure, production, indicators, and CO2 emissions estimates. Retrieved on Oct 29, 2026 from https://www.eia.gov/state/seds/seds-data-complete.php#CompleteDataFile.

BEA: Gross Domestic Product by State and Industry (SAGDP2)

- **Provides:** State GDP by industry in current dollars, used to compute the industrial share of the economy: (mining + manufacturing) ÷ all-industry GDP. Current dollars are used because chained (real) dollars do not add correctly across industries.
- **Citation:** U.S. Bureau of Economic Analysis. Gross Domestic Product by State and Industry (SAGDP2), 1997–2025. Retrieved on Nov 2, 2026 from https://apps.bea.gov/regional/downloadzip.htm.

USGS: Construction Aggregates

- **Provides:** Annual state-level production of construction sand and gravel and crushed stone (metric tons), used as a measure of construction activity. Construction sand and gravel is the variable used, because USGS withholds crushed stone figures for Delaware and Louisiana in most years.
- **Citation:** U.S. Geological Survey (USGS) National Minerals Information Center, Aggregates Time Series Data by State, Type, and End Use, 1971–2023. Retrieved on Oct 19, 2026 from https://www.usgs.gov/media/files/usgs-aggregates-time-series-data-state-type-and-end-use.

U.S. Forest Service: Forest Inventory and Analysis (FIA) ***(robustness only)***

- **Provides:** State-level forest area change, carbon density, and timber harvest.
- **Citation:** U.S. Department of Agriculture, Forest Service. (2024). Forest Inventory and Analysis EVALIDATOR (Version 2.0) [Web application]. U.S. Department of Agriculture, Forest Service, Forest Inventory and Analysis Program. Retrieved on Oct 19, 2026 from https://research.fs.usda.gov/products/dataandtools/evalidator-and-fiadb-api.

Feature Set B: Citizen and Consumer Demand Metrics

FHWA: Highway Statistics Series

- **Provides:** Annual vehicle-miles traveled (VMT) by state, rural and urban, converted to VMT per person.
- **Citation:** U.S. Department of Transportation. (2024). Federal Highway Administration, Office of Highway Policy Information. Highway Statistics Series. Retrieved on Oct 29, 2026 from https://www.fhwa.dot.gov/policyinformation/statistics.cfm.

BTS: Air Carrier Statistics (T-100 Segment Data)

- **Provides:** Domestic air passenger-miles (passengers × segment distance) and freight ton-miles, by state of departure.
- **Citation:** U.S. Department of Transportation. (2026). Bureau of Transportation Statistics. Air Carriers: T-100 Segment (All Carriers). Retrieved on Oct 29, 2026 from https://www.transtats.bts.gov/Fields.asp?gnoyr_VQ=FIM.

EIA: State Energy Data System (Residential Electricity)

- **Provides:** Residential electricity purchases per person, and electricity sales by sector, used to split electric power emissions between households and businesses in the descriptive analysis.
- **Citation:** Same SEDS file as Feature Set A.

BEA: Personal Consumption Expenditures by State (SAPCE2)

- **Provides:** Personal consumption expenditures per person by type of product, used as a proxy for consumer demand. Spending on gasoline and other energy goods is subtracted, because that spending is fuel use.
- **Citation:** U.S. Bureau of Economic Analysis. Per Capita Personal Consumption Expenditures by State (SAPCE2), 1997–2025. Retrieved on Nov 2, 2026 from https://apps.bea.gov/regional/downloadzip.htm.

USDA National Agricultural Statistics Service (NASS)

- **Provides:** State annual livestock inventories from the NASS survey program (cattle and calves, milk cows, hogs, sheep), used to compute cattle per person. Livestock is the largest source of agricultural methane.
- **Citation:** U.S. Department of Agriculture. (2026). National Agriculture Statistics Service, Data & Statistics: Quick Stats Large Datasets. Retrieved on Oct 29, 2026 from https://www.nass.usda.gov/datasets/.

Spatial Weights

U.S. Census Bureau: TIGER/Line Shapefiles, States (2022)

- **Provides:** State boundary polygons, used to build a queen-contiguity neighbor matrix for the 50 states (DC and territories dropped). Queen contiguity counts shared corners, so the Four Corners pairs (AZ–CO, UT–NM) are neighbors. The matrix is row-standardized and saved as an edge list in data/state_neighbors.csv (state_fips, neighbor_fips, weight): 110 neighbor pairs, 220 rows. build_panel.py uses it to add each index's neighbor average in the same year (W_D_corporate, W_D_household, and the robustness indices).
- **Limitations:** Alaska and Hawaii have no neighbors, so their spatial lag is set to 0 (flag has_neighbors = 0). Neither is in the main sample (no degree days), so this matters only for a 50-state specification without HDD/CDD. TIGER/Line state polygons include Great Lakes and coastal water, so three links cross water only: IL–MI, MI–MN and NY–RI. They are kept.
- **Citation:** U.S. Census Bureau. (2022). TIGER/Line Shapefiles: States (tl_2022_us_state). Retrieved on Nov 2, 2026 from https://www2.census.gov/geo/tiger/TIGER2022/STATE/.

Reproducing the Data

1. Clone the repository. The cleaned data in data/interim/ and data/processed/ is already included, so **steps 2–5 are only needed to rebuild it from the original sources**.
2. Download each source from the links above into the matching data/raw/ subfolder (data/raw/outcome/, data/raw/climate/, data/raw/corporate/, data/raw/citizen/). File names and retrieval dates are listed with each citation.
3. Run each loader in code/data_loaders/. Each one takes the raw file or folder, data/state_crosswalk.csv, and an output folder, and writes one tidy file to data/interim/:

Loader	Writes
load_epa.py	data/interim/outcome/epa.csv
load_eia.py	data/interim/shared/eia.csv
load_bts.py	data/interim/shared/bts.csv
load_bea.py	data/interim/shared/bea.csv
load_noaa.py	data/interim/shared/noaa.csv
load_fhwa.py	data/interim/civilian/fhwa.csv
load_usda.py	data/interim/corporate/usda.csv
load_usgs_minerals.py	data/interim/corporate/usgs_minerals.csv

Example: `python3 code/data_loaders/load_epa.py data/raw/outcome/allstateghgdata90-22_v082924/Al data/state_crosswalk.csv data/interim/outcome`

4. Download tl_2022_us_state.zip (see Spatial Weights) into data/raw/shared/TIGER/ and build the neighbor matrix: `python3 code/data_preparation/build_spatial_weights.py data/raw/shared/TIGER/tl_2022_us_state.zip data/state_crosswalk.csv data/state_neighbors.c` The output is committed, so this step is only needed to rebuild it.

5. Run build_panel.py (`python3 code/data_preparation/build_panel.py data/interim data/state_crosswalk.csv data/state_neighbors.csv data/processed`) to join every interim file into data/processed/panel.csv (one row per state-year, 50 states × 1997–2022 = 1,300 rows) and data/processed/data_dictionary.csv. It prints a report with coverage checks, missingness, and driver correlations.
6. Run the Phase 3 decomposition and EDA: `python3 code/data_preparation/decompose_emissions.py data/processed/panel.csv data/processed/decomposition.csv` `python3 code/analysis/eda.py data/processed/panel.csv data/processed/decomposition.csv visuals data/processed`
7. Run the Phase 4 fixed-effects baseline: `python3 code/models/fe_baseline.py data/processed/panel.cs models findings/baseline`. Each specification (main, no_gdp, no_wy, no_ippu) gets its own folder under models/fe_baseline/ with the fitted model (model.pkl), coefficients.csv, metrics.csv and summary.txt. The combined comparison table is models/fe_baseline/table.md; the gap test (D_corporate − D_household) and the normal vs. t(G − 1) comparison go to findings/baseline/.
8. data/raw/ is git-ignored, so downloaded data will not be committed.